Practical# 05

Setup a continues integration CI pipeline using GitHub Action that automatically builds and pushes a Docker image to Docker Hub whenever code is pushed. Created a workflow file that checks out the repo, logs in the Docker Hub using GitHub secrets builds the image from the Dockerfile, tags it as username/app-ci:latest, and pushes it. Also briefly explain the role of environment variable and secrets in workflow.

```
Command Prompt

C:\Users\narut\docker-ci-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\docker-ci-app

21-04-2026  12:35    <DIR>          .
21-04-2026  12:33    <DIR>          ..
21-04-2026  12:34               195 app.py
21-04-2026  12:35               115 Dockerfile.txt
21-04-2026  12:34                 5 requirements.txt
               3 File(s)            315 bytes
               2 Dir(s)   6,762,741,760 bytes free

C:\Users\narut\docker-ci-app>mkdir .github\workflows

C:\Users\narut\docker-ci-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 2461-2C7D

 Directory of C:\Users\narut\docker-ci-app

22-04-2026  21:04    <DIR>          .
21-04-2026  12:33    <DIR>          ..
22-04-2026  21:04    <DIR>          .github
21-04-2026  12:34               195 app.py
21-04-2026  12:35               115 Dockerfile.txt
21-04-2026  12:34                 5 requirements.txt
               3 File(s)            315 bytes
               3 Dir(s)   6,763,266,048 bytes free

C:\Users\narut\docker-ci-app>cd .github

C:\Users\narut\docker-ci-app\.github>cd workflows

C:\Users\narut\docker-ci-app\.github\workflows>notepad ci-cd.yml

C:\Users\narut\docker-ci-app\.github\workflows>
```

STEP-01: Create directory and get inside the directory.

STEP-02: Create the file and the content of app.py, Dockerfile, requiremet.txt

STEP-03: write the yml file in the dir .github/workflows/

STEP-04: Create the repository

STEP-05: go to settings >> secrets and variable >> actions >> add secretes

- Secret 1

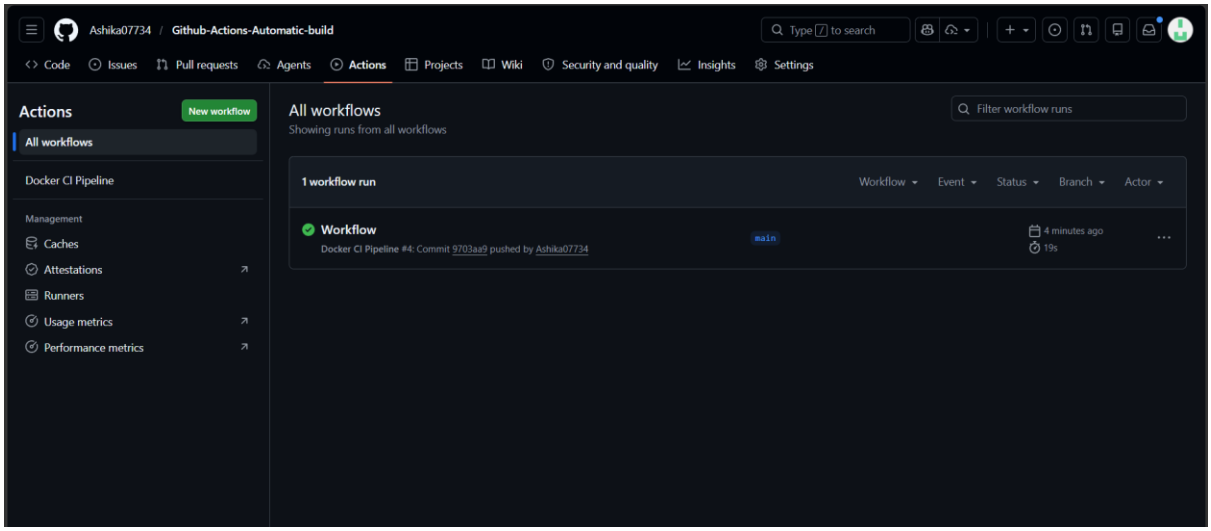
Name: DOCKER_USERNAME
Value: your-dockerhub-username

- Secret 2

Name: DOCKER_PASSWORD
Value: your-dockerhub-password

```
User@Ashika MINGW64 ~ (master)
$ cd docker-ci-app

User@Ashika MINGW64 ~/docker-ci-app (master)
$ git init
Initialized empty Git repository in C:/Users/narut/docker-ci-app/.git/

User@Ashika MINGW64 ~/docker-ci-app (master)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
        add

User@Ashika MINGW64 ~/docker-ci-app (master)
$ git add .

User@Ashika MINGW64 ~/docker-ci-app (master)
$ git commit -m "ci-cd pipeline"
[master (root-commit) af8dce2] ci-cd pipeline
 4 files changed, 20 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 Dockerfile.txt
 create mode 100644 app.py
 create mode 100644 requirements.txt

User@Ashika MINGW64 ~/docker-ci-app (master)
$ git branch -M main

User@Ashika MINGW64 ~/docker-ci-app (main)
$ git remote add origin https://github.com/Ashika07734/Github-Actions-Automatic-
build.git

User@Ashika MINGW64 ~/docker-ci-app (main)
$ git push -u main
```

STEP-06: Intialize, add, commit the push



STEP-07: Workflow of the project